Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Michael Sealy	Email	coria-consortium@purdue.edu
Institute Name	Purdue University			
Address	585 Purdue Mall, West Lafayette, IN 47907-2088			

2. Technology Overview

Title of Proposed Technology	**Multimodal AI-based State Estimation and Anomaly Detection for Autonomous Manufacturing System**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and Materials		*Information and Communication Technology (ICT)*	
	Medium Category	Small Category	Medium Category	Small Category
	Additive Manufacturing Equipment	*Binder Jet Additive Manufacturing Systems*	*Additive Manufacturing Equipment*	*Binder Jet Additive Manufacturing Systems*
Concept of Proposed Technology	**Core Concept** Binder Jet additive manufacturing is a representative high-risk and non-observable manufacturing process in which powder spreading, binder jetting, and bonding behaviors interact probabilistically across long-duration builds. This project proposes an AI-based process reliability framework that enables in-process state awareness and predictive maintenance through multimodal sensing and state estimation. **Key Functions and Potential Applications** ● The proposed technology integrates multimodal sensing (vision, sound, and current) to indirectly observe process behavior and capture subtle anomaly signatures that are not detectable by single-sensor approaches. ● An AI-based state estimation model is applied to infer hidden process states from multimodal			

data, enabling early detection of process degradation and anomaly progression rather than simple threshold-based alarms.

- Based on estimated process states, predictive maintenance and operational decision logic are implemented to prevent build failures and support autonomous manufacturing operation.

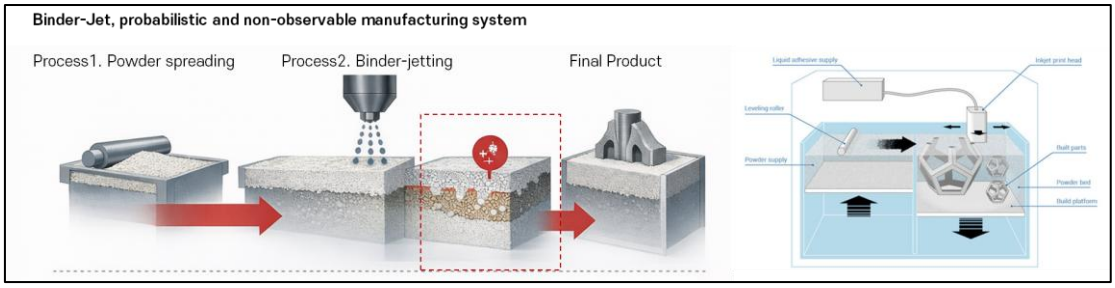

Fig. 1. Target Manufacturing System & Engineering Challenge

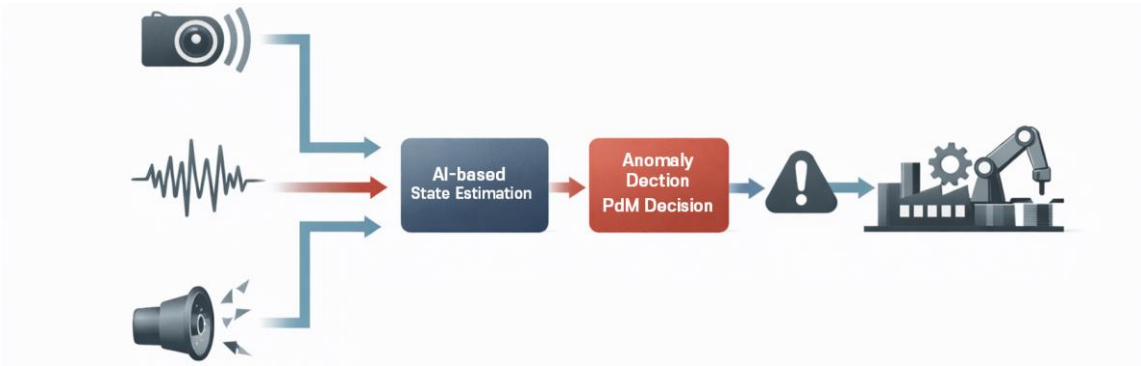

Fig. 2. Multimodal Sensing–State Estimation–PdM Architecture

Rationale for Joint R&D Project

Critical Technical Barriers

- Binder Jet is selected as the target system due to its long build time, high sensitivity to micro-process anomalies, and strong dependence on post-build inspection, which collectively result in high risk of large-scale build failures.
- Key process quality determinants (e.g., binder infiltration, local bonding stability, density distribution) cannot be directly measured during printing, making conventional sensor-based monitoring insufficient.

Why Global Collaboration?

Binder Jet additive manufacturing presents structural limitations in process observability and reliability, which cannot be effectively addressed through conventional sensor-based monitoring or single-organization development efforts.

Accordingly, this project adopts an international joint R&D framework to establish a validated AI-based process reliability technology applicable to industrial manufacturing environments.

1) Existing sensing implementations in Binder Jet systems are limited to partial signal observation and are not structured to support quantitative defect judgment or quality

	prediction during production. 2) Conventional approaches rely on isolated sensor signals and threshold-based alarms, lacking explicit linkage between sensor data and actual process quality or failure risk. 3) In probabilistic and non-observable processes, micro-anomalies accumulate gradually, making it difficult to assess anomaly progression and build failure risk without state estimation–based interpretation. 4) Development of such technology requires long-term expertise in sensing, signal interpretation, and process modeling, which exceeds the capability of a single SME-led effort. 5) Research institute contributes fundamental research capabilities in high-resolution sensing, multimodal signal analysis, and state estimation algorithms, forming the theoretical backbone of the proposed AI framework. 6) KITECH provides process-level validation expertise, including accelerated and stress testing, ensuring reproducibility, reliability, and industrial applicability of the developed technology. 7) Korean SME partner, as a domestic Binder Jet system developer and operator, enables direct access to equipment-level data, real operating conditions, and rapid field validation, which are structurally inaccessible in globally closed equipment ecosystems. 8) This consortium structure enables the integration of fundamental research, industrial validation, and commercialization pathways, justifying the necessity of a government-supported international joint R&D program.
Key Words	1. Binder Jet Additive Manufacturing 2. Multimodal Process Sensing 3. AI-based State Estimation 4. Predictive Maintenance (PdM) 5. Autonomous Manufacturing Reliability

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)

R&D Objectives

R&D Objectives

The ultimate objective of this project is to establish and validate an AI-based process reliability platform for industrial Binder Jet additive manufacturing by securing large-scale multimodal process data and implementing state estimation–driven predictive maintenance technologies.

Unlike small-scale proof-of-concept studies, this project aims to achieve industrial applicability through statistically reliable datasets and repeated validation under accelerated and stress operating conditions.

Detailed Objectives

1) Objective 1: Large-scale Multimodal Process Data Infrastructure
   - Establish a structured data acquisition and management framework by collecting more than **500,000 multimodal process data instances** (vision, sound, current) from continuous and repetitive Binder Jet builds, enabling statistically meaningful AI training and validation.
2) Objective 2: Multimodal Process Sensing Framework
   - Develop a Binder Jet–optimized sensing configuration that enables early awareness of subtle process anomalies through correlation analysis across vision, acoustic, and current signals, overcoming the limitations of single-sensor monitoring.
3) Objective 3: AI-based State Estimation and Anomaly Detection
   - Develop AI models that estimate hidden process state variables and identify anomaly progression trends, enabling interpretation of process degradation rather than binary anomaly alarms.
4) Objective 4: Accelerated and Stress Test-based PdM Validation
   - Design and implement an accelerated and stress testing environment that intentionally reproduces process anomalies, ensuring reproducibility, reliability, and quantitative validation of predictive maintenance performance.
5) Objective 5: PdM-driven Operational Decision and Autonomous Manufacturing Linkage
   - Integrate AI-based state estimation results into practical operational decision logic, supporting early failure prevention, maintenance planning, and reduced human dependency in long-duration manufacturing operations.

R&D Target Specifications

Key Performance Indicator	Target	Weight	Measurement and Verification Method
Process Data Volume	≥ 500,000 data points	25%	Verification of cumulative data volume based on multimodal sensor data logs (including

			vision, sound, and current data)
Anomaly Detection Accuracy	≥ 90%	20%	Analysis of the success rate of prior detection relative to actual anomaly occurrence events during accelerated and stress condition tests
Pre-failure Detection Rate (Build Failure)	≥ 85%	20%	Analysis of whether risk warnings are generated at stages prior to build completion during failed builds
PdM Lead Time	≥ 30% earlier detection	15%	Comparison of the time difference between actual process anomaly occurrence and AI alert generation timing
Stress Test Reproducibility	≥ 90%	10%	Evaluation of consistency in sensor signals and AI judgment results under repeated tests with identical stress conditions
State Estimation Model Accuracy	≥ 85%	10%	Number of independently trained and validated state estimation and anomaly detection models corresponding to different process anomaly scenarios, evaluated under accelerated and stress test conditions

R&D Plan

Research Approach & Strategy

This project adopts an integrated research strategy that connects process sensing, AI-based state estimation, and operational decision-making into a unified process reliability framework for industrial Binder Jet additive manufacturing.

1) Integrated Process–Data–Operation Strategy
   - A unified AI-PdM platform is established by linking Binder Jet process data acquisition, state estimation–based AI modeling, and predictive maintenance decision logic into a single data flow.
2) Large-scale Data and Accelerated Validation Strategy
   - Rather than relying on small-scale experiments, the project secures more than 500,000 multimodal process data instances and repeatedly reproduces process anomalies using accelerated and stress testing environments to ensure statistical reliability and reproducibility of PdM performance.
3) Demonstration-oriented Industrial Application Strategy
   - From the early research stage, real manufacturing scenarios—such as long-duration builds, repetitive production, and delivery-sensitive operations—are explicitly reflected, targeting field-applicable AI-based process reliability technologies rather than laboratory-level algorithm validation.

Key Tasks & Milestones

Year	Core Objective	Major Tasks	Expected Outcomes

			(Milestones)
Year 1	Establishment of Data and Testing Infrastructure	- Installation of multimodal sensors (vision, sound, current) - Construction of process data structuring and acquisition pipeline - Design and fabrication of accelerated and stress testing equipment (Test Rig) - Definition of initial process anomaly scenarios	- Acquisition of more than 200k instances of process data - Completion of one accelerated and stress testing device
Year 2	Development and Verification of AI Models	- Development of AI-based process state estimation models - Establishment of structures for distinguishing normal, boundary, and abnormal states - Performance verification of anomaly detection based on accelerated testing	- Cumulative acquisition of more than 500k instances of data - Achievement of ≥ 90% early anomaly detection accuracy
Year 3	PdM Demonstration and Securing Applicability	- Demonstration of PdM-based process operation scenarios - Verification under long-duration builds and repeated production environments - Establishment of business expansion and commercialization strategies	- Achievement of ≥ 85% pre-failure detection rate for build failures - Completion of AI-PdM platform demonstration

Methodology

1) [Sensing] Multimodal Process Signal Acquisition:
   - During the Binder Jet additive manufacturing process, multimodal sensors—including vision, sound, and current—are used to collect powder spreading behavior, binder jetting behavior, and process load variations in real time, enabling indirect observation of process states
2) [Data] Large-scale Process Data Structuring and Management:
   - Process data are structured at layer-level, time-level, and event-level, and a cumulative dataset of more than 500,000 instances—including normal, boundary, and abnormal states—is constructed for AI training and validation
3) [AI] State Estimation-based Anomaly Detection:
   - AI models that estimate hidden process states (state variables) using multimodal sensor data as inputs are applied to enable stepwise assessment of process degradation and anomaly progression, rather than simple threshold-based judgment
4) [Validation] PdM Performance Verification Based on Accelerated and Stress Condition Testing:
   - Accelerated and stress condition testing equipment is utilized to repeatedly

	reproduce process anomalies, and the early anomaly detection performance of AI models and predictive maintenance (PdM) alert lead time are quantitatively verified 5) [Operation] Integration of PdM-based Process Decision-making: ● Based on AI state estimation results, predictive maintenance decision-making logic applicable to actual equipment operation—such as nozzle inspection, process condition adjustment, and build interruption decisions—is implemented

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	This project assigns clearly differentiated roles to each research institute to ensure effective integration of fundamental research, process validation, and industrial applicability. **Research Institute (Fundamental Technology Development)** Research institute is responsible for developing core technologies related to multimodal sensing, signal interpretation, and AI-based state estimation. ● This includes sensor configuration design, high-resolution signal analysis, multimodal data fusion, and development of state estimation algorithms that infer hidden process states and anomaly progression. **Korea Institute of Industrial Technology (KITECH) (Process Validation and Verification)** KITECH is responsible for designing and operating accelerated and stress testing environments that intentionally reproduce Binder Jet process anomalies. ● KITECH establishes quantitative PdM validation criteria, conducts repeated verification tests to ensure reproducibility and reliability, and supports standardization of validation methodologies for industrial application.
Expected R&R of Korean SME Partner	Korean SME partner serves as the industrial execution and commercialization anchor of the project by providing real equipment environments, process data, and application pathways. ● Korean SME partner provides industrial Binder Jet additive manufacturing systems and real operating environments required for large-scale data acquisition and field validation. ● The company is responsible for stable collection, management, and preprocessing of multimodal process data generated during actual equipment operation. ● AI-based state estimation and PdM models developed by research partners are integrated into the equipment control environment for performance demonstration and validation. ● Based on validation results, Korean SME partner refines operational conditions and decision logic, and leads productization and commercialization of AI-based PdM functions.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This project is expected to establish a validated AI-based process reliability technology for industrial additive manufacturing, with direct impact on manufacturing quality, operational stability, and commercialization potential. **Technical Outcomes** ● Establishment of an AI-based process monitoring and state awareness framework for Binder Jet additive manufacturing by integrating large-scale multimodal process data. ● Early detection and prevention of build failures through state estimation–driven predictive maintenance, reducing reliance on post-build inspection. ● Accumulation of a statistically reliable dataset exceeding 500,000 process data instances, forming a high-entry-barrier technical asset that is difficult to replicate in the short term. **Commercialization Potential** ● Integration of AI-based PdM functions as standard or optional features in industrial Binder Jet systems, providing clear technological differentiation. ● Improvement of equipment utilization and reduction of build failure rates, enabling quantifiable value propositions for customers in aerospace, energy, automotive, and shipbuilding industries. ● Creation of sustainable revenue models through software features, maintenance services, and data-driven upgrades beyond one-time equipment sales. **Industrial and National Impact** ● Fundamental improvement of manufacturing cost structures by reducing material loss, rework, and delivery delays caused by build failures. ● Strengthening the global competitiveness of domestic SMEs by demonstrating successful application of AI manufacturing technologies on real industrial equipment. ● Contribution to national strategic capabilities by accelerating the transition toward autonomous manufacturing through AI-based process reliability technologies applicable across multiple manufacturing domains.

6. References

(1) King, W. E., Barth, H. D., Castillo, V. M., et al.; Observation of keyhole-mode laser melting in laser powder-bed fusion; Journal of Materials Processing Technology, Vol. 214, No. 12, 2014, pp. 2915–2925; DOI: 10.1016/j.jmatprotec.2014.06.005

(2) Grasso, M., & Colosimo, B. M.; Process defects and in situ monitoring methods in metal powder bed fusion; Measurement Science and Technology, Vol. 28, No. 4, 2017, Article 044005; DOI: 10.1088/1361-6501/aa5c4f

(3) Zonta, T., da Costa, C. A., da Rosa Righi, R., et al.; Predictive maintenance in the Industry 4.0: A systematic literature review; Computers & Industrial Engineering, Vol. 150, 2020, Article 106889; DOI: 10.1016/j.cie.2020.106889

(max 10 pages)

(4) Lee, J., Bagheri, B., & Kao, H. A.; A cyber-physical systems architecture for Industry 4.0-based manufacturing systems; Manufacturing Letters, Vol. 3, 2015, pp. 18–23

(5) Sim, Y., Kim, E., Lee, H., Sohn, J., & Jun, M. B. G.; Structured sound data augmentation based on spectral analysis for machine tool operational state monitoring; The International Journal of Advanced Manufacturing Technology, Vol. 141, 2025, pp. 1705–1719; DOI: 10.1007/s00170-025-16797-w.

(6) Han, C., Kang, Y. S., Sim, Y., Park, H. W., & Jun, M. B. G.; LISTEN: Lightweight Industrial Sound-representable Transformer for Edge Notification; arXiv preprint arXiv:2507.07879, 2025.

(7) Kim, E., Bui, T., Yuan, J., Mouli, S. C., Ribeiro, B., Yeh, R. A., Fassnacht, M. P., & Jun, M. B. G.; Online real-time machining chatter sound detection using convolutional neural network by adopting expert knowledge; Manufacturing Letters, Vol. 41, 2024, pp. 1386–1397; DOI: 10.1016/j.mfglet.2024.09.165.

(8) Kim, E., Jeon, J., Kim, Y., Yun, H., Wellman, J., Choi, Y. W., Lee, S. W., Jun, M. B. G., & Lee, J.; Control-Resilient Roller Wear Prediction for Thin Wire Flattening Process via an Internal Sound-Guided Dynamic Conditional Network; International Journal of Precision Engineering and Manufacturing - Green Technology, Vol. 12, No. 3, 2025, pp. 919–934; DOI: 10.1007/s40684-025-00712-5.